

Can controlled phase offsets synchronize?

exp085 · 18 August 2026 · Draft

Abstract

Two mature, slightly detuned PING circuits begin at four prescribed relative phases before reciprocal excitation switches on. The maximum onset-phase error is 0.0004 rad. With $K = 0.08$, terminal phase error falls to a median 0.14 rad; matched uncoupled controls retain 1.45 rad error. Controlling onset phase turns an accidental representative trajectory into a direct test of convergence.

Methods

1. **Scout mature uncoupled dynamics.** Independent 100 Hz/channel Poisson streams drive two 80 E / 20 I PING circuits. Circuit A uses 4 ms inhibitory decay and circuit B uses 5 ms. Each seed runs uncoupled for 2400 ms.
2. **Select valid phase-controlled states.** Smoothed E-population rates are filtered over 20–60 Hz. The runner locates naturally occurring joint states nearest the four prescribed phases. It then deterministically replays the uncoupled prefix and captures the complete graph runtime state at each onset. No voltage, conductance, refractory counter, population history, or delay history is edited or recombined.
3. **Apply paired futures.** Each captured state branches into $K = 0.08$ coupling-on and $K = 0$ controls. Within a seed, every prescribed phase receives the same future private A/B input streams. Exact replay assertions compare every population spike against the scout prefix before any branch is accepted.
4. **Measure convergence.** Each trajectory is compared with its own mean relative phase over the final 500 ms. Circular terminal-phase error is smoothed over 50 ms. The final 100 ms are excluded uniformly as a filter-edge guard.

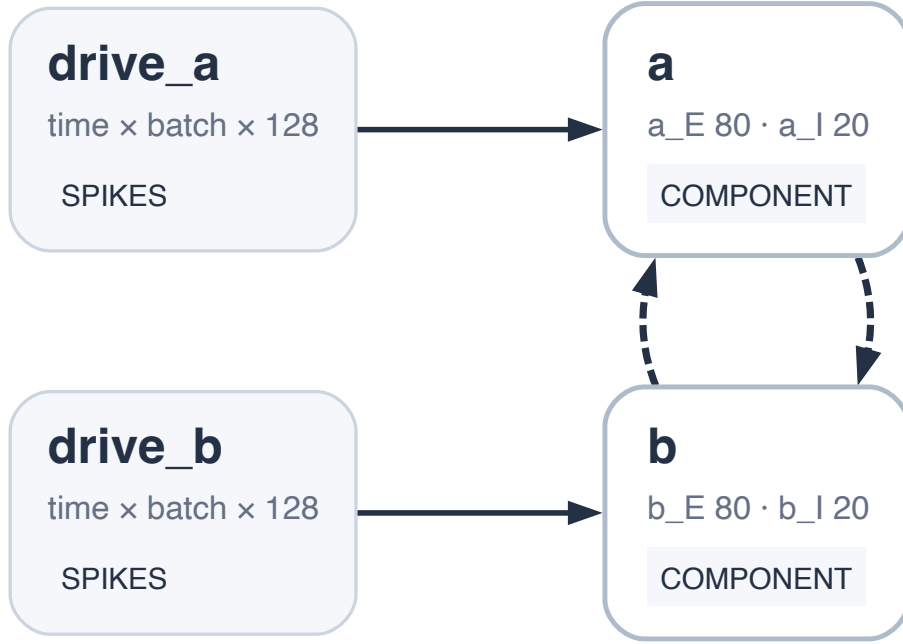


Figure 1: Controlled coupling-onset graph. Independent spike inputs drive circuits A and B. Four reciprocal AMPA projections switch from zero during scout and replay to $K = 0.08$ after each captured onset.

Results

The prescribed offsets are recovered to sub-milliradian accuracy across all 20 onsets.

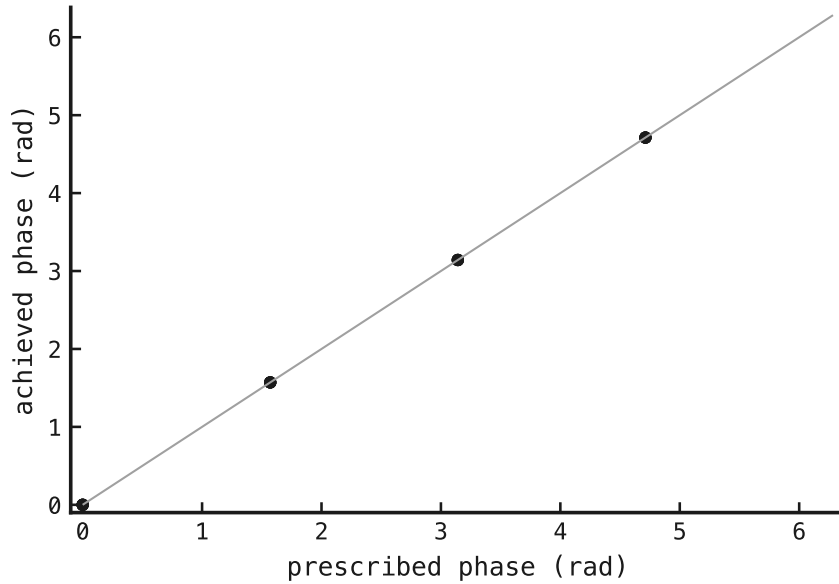


Figure 2: Phase-control accuracy. Each point is one seed and prescribed onset phase; the diagonal is exact agreement. Maximum circular target error is 0.00044 rad.

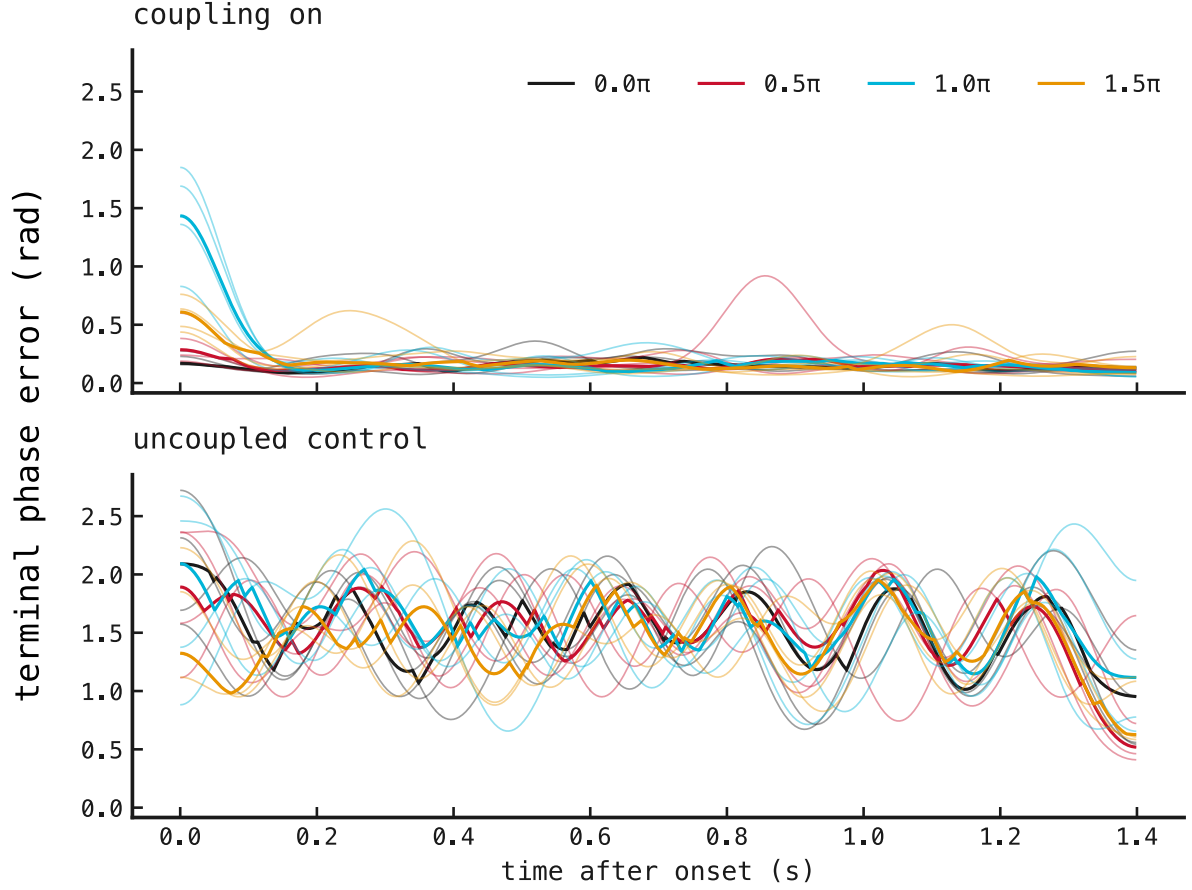


Figure 3: Convergence from controlled relative phases. Top: coupling switches to $K = 0.08$. Bottom: matched states continue uncoupled. Thin traces are all 5 seeds at each prescribed phase; thick traces are phase-group medians. Coupled trajectories collapse toward low terminal error, whereas uncoupled controls continue to traverse relative phase.

Controlled onset resolves the ambiguity in the earlier accidental-phase experiment: convergence is a repeatable consequence of reciprocal coupling here, not a fortunate single seed. The terminal reference remains retrospective, so the figure demonstrates convergence without registering a synchronization-latency estimator.